Module 7: Classification

Part 1: Logistic Regression

Wenting Xu

Iowa State University

Copyright and Attribution

Some lecture slides and instructional materials in this course are adapted from the following sources:

- *[1] https://blog.alliedoffsets.com/understanding-the-difference-between-linear-and-logistic-regression-models*

Example of classification problems

- Goal: carry out classification of a response Y using predictors $X_1, \dots, X_p$.

- Main difference with the regression models we've covered so far is that now Y is qualitative (categorical).

- Still in the supervised learning setting. Classification is NOT clustering!!

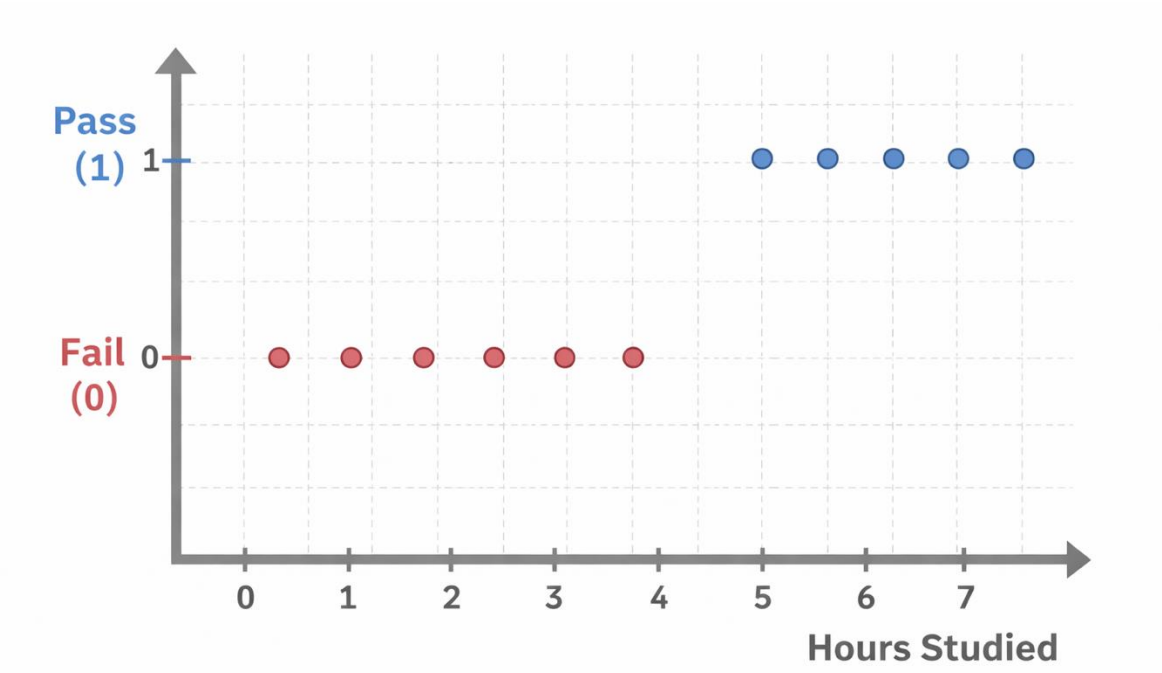
Example of classification problems

- Email is spam or not spam?
- Is this transaction a fraud or not fraud?
- Does an individual have a disease or not a disease? Is this DNA mutation harmful or not?
- Is this image of a dog or not a dog (image classification)?
- ....

Example

Suppose we are trying to classify the test result of students:

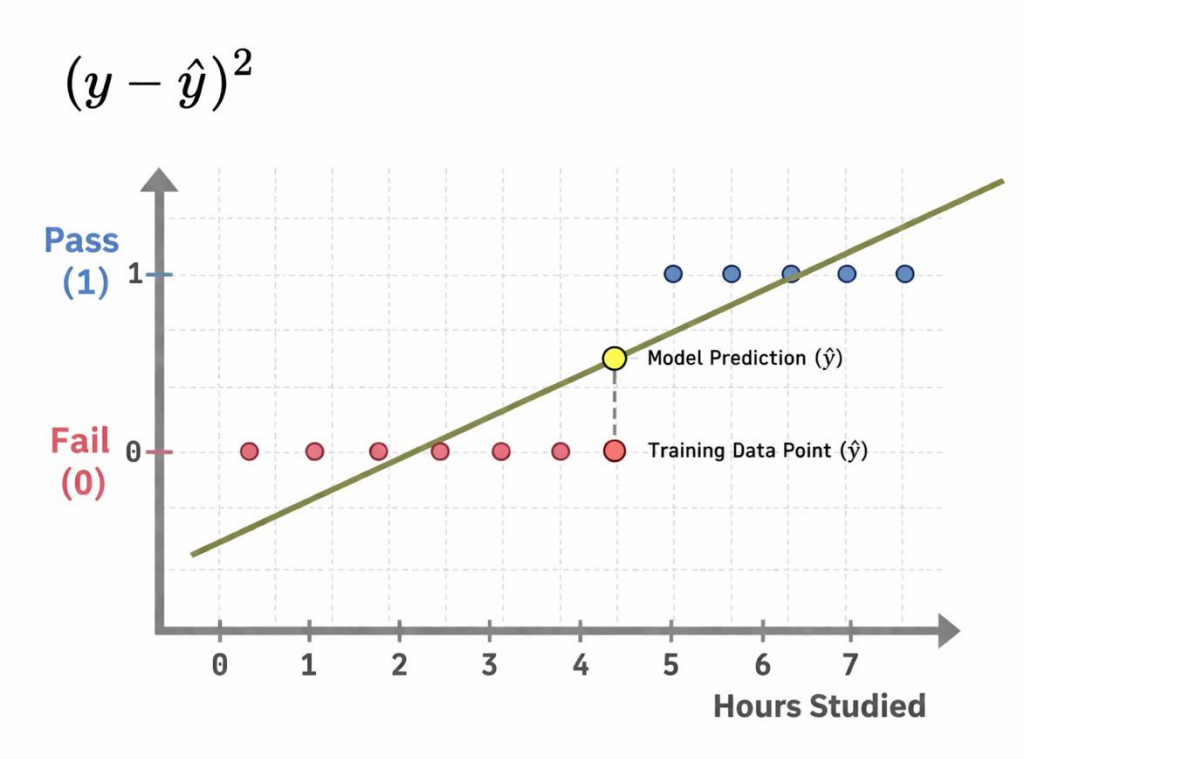
$$Y = \begin{cases} 1 & \text{if } pass \\ 0 & \text{if } fail \end{cases}$$



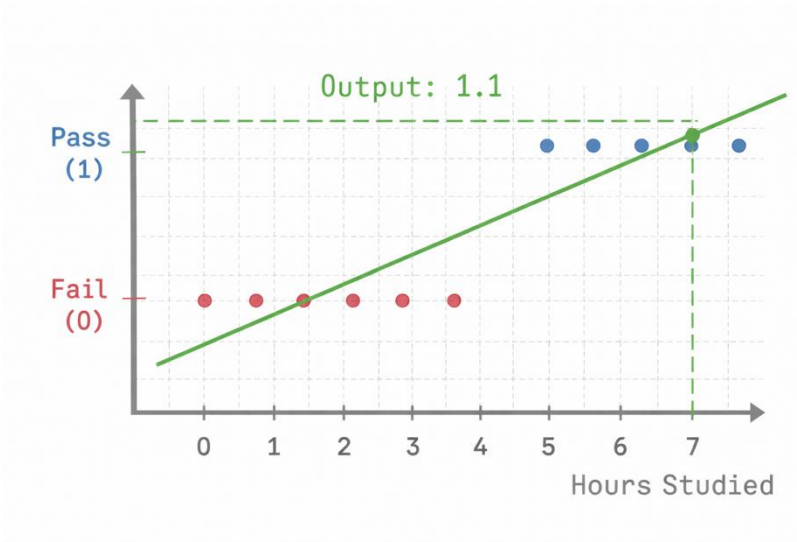
Example

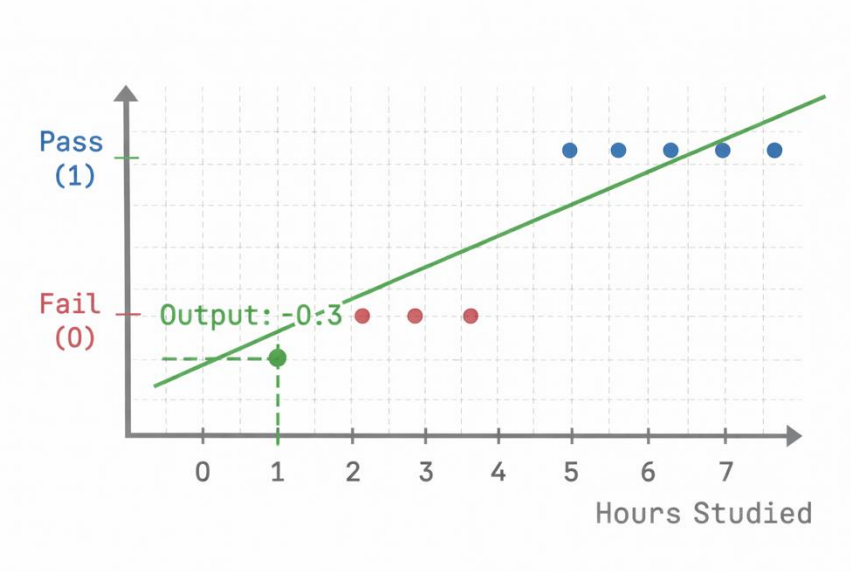
Why don't we use a least squares model here to predict Y ?

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 * number\ of\ hours\ studied$$



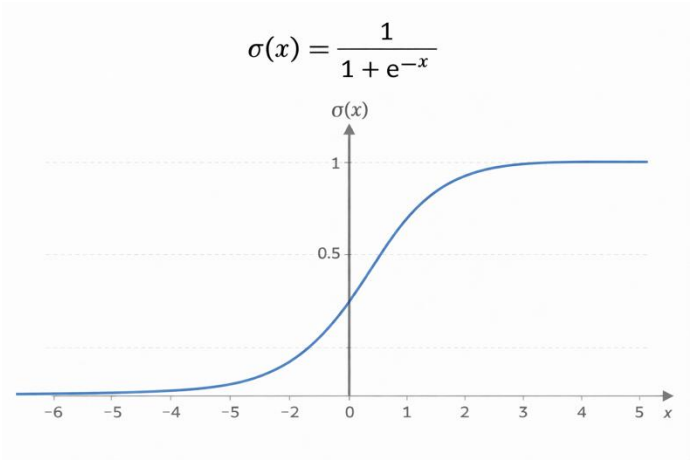
Example

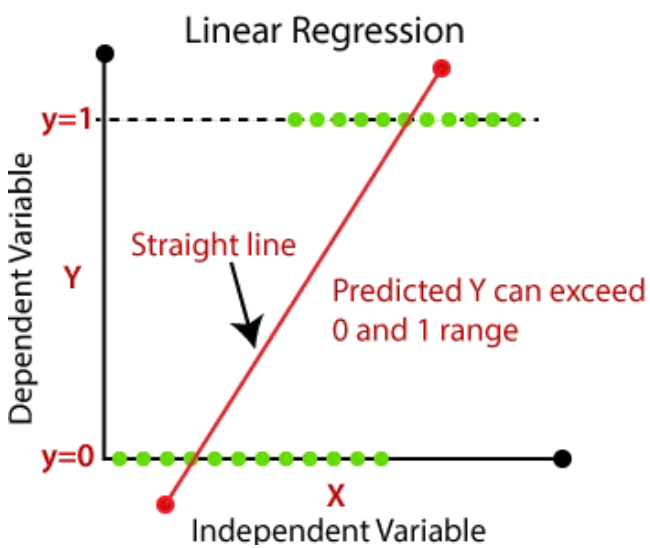


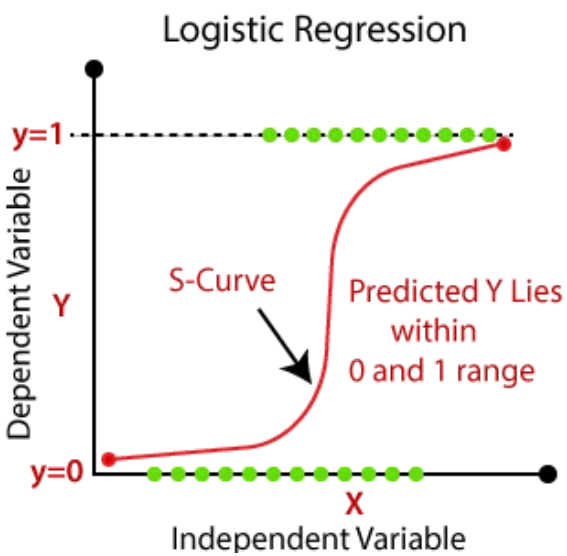


- In classification, we usually want something interpretable as a probability (between 0 and 1).
- Linear regression can give values **< 0 or > 1**, which don't make sense as probabilities.

Sigmoid function







Estimating beta

Instead we need to use classification techniques that are specifically designed to handle this kind of problem:

1. **Logistic regression**
2. Linear/quadratic discriminant analysis
3. Naive Bayes
4. k-nearest neighbor

Gold standard for classification is **Bayes Rule**.

Bayes Rule

Let's start with a Y that only takes 2 values:

$$Y = \begin{cases} 1 & \text{if } pass \\ 0 & \text{if } fail \end{cases}$$

Target: $P\left(Y = 1 \mid X_1, \dots, X_p\right)$.

If I knew this probability, I could use Bayes Rule directly:

$$\begin{cases} P(Y = 1 \mid X) > 0.5 & \text{then } \hat{Y} = 1 \\ P(Y = 1 \mid X) \leq 0.5 & \text{then } \hat{Y} = 0 \end{cases}$$

Logistic Regression

Instead of modeling Y directly, let's try to model the probability that Y belongs to a category.
Suppose Y only takes 2 values:

$$Y = \begin{cases} 1 & \text{if } pass \\ 0 & \text{if } fail \end{cases}$$

Target: $P(Y = 1 \mid X_1, \dots, X_p) = p(x)$.

- $p(x)$ is usually unknown to us, so we need to estimate it from the data.
- We must model $p(x)$ using a function that can guarantee outputs fall between 0 and 1 .

In logistic regression, we use the logistic function:

$$p(x) = \frac{\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p)}$$

Estimating β

$\beta_0, \beta_1, \ldots, \beta_p$ are parameters. These are unknown quantities and we need to estimate them from our data $\Rightarrow$ **Maximum likelihood estimation.**

- Intuition: We seek estimate for $\beta_0, \beta_1, \ldots, \beta_p$ such that the predicted probability $\hat{p}(x) = \hat{P}(Y = 1 \mid X)$ is large for observations where $Y = 1$ and small for observations where $Y = 0$.
- Following our example, that means
- If an individual student passed the exam($Y = 1$), then we want $\hat{p}(x)$ to be close to 1 .
- If an individual student failed the exam ($Y = 0$), then we want $\hat{p}(x)$ to be close to 0 .

Maximum likelihood estimation technical details (sort of)

R code

See R script: logit_intro.R